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ORIGINAL RESEARCH

Modern Cardiac ICU Care Delivery and the Role of the Cardiac ICU Cardiologist

Submitted on behalf of the American College of Cardiology's Critical Care Cardiology Council and the Society of Critical Care Cardiology

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ABSTRACT

BACKGROUND The cardiac intensive care unit (CICU) has evolved into a complex care environment for critically ill patients with cardiac and noncardiac diseases.

OBJECTIVES Our goal was to describe contemporary CICU care delivery and the role of cardiologists therein.

METHODS The American College of Cardiology administered a 42-item survey to U.S. and Canadian CICU-focused cardiologists designed to capture models of care delivery and workforce demographics.

RESULTS The survey was distributed by email to 1,085 U.S. and Canadian CICU cardiologists. The response rate was 20%, yielding a final sample of 166 after excluding trainees and those not board-certified or board-eligible in cardiology. The majority were from medium (34%) or large (64%) academic (81%) medical centers. Fifty-three percent reported working in high-intensity care models and 61% reported that their CICU was dedicated exclusively to medical cardiology patients. Critical care medicine-boarded physicians contributed to care through consultative (53%), comanagement (29%), and/or primary roles (44%). Subspecialization beyond cardiology was common (82%), with critical care medicine being most frequent (46%), followed by echocardiography (37%), advanced heart failure (21%), and interventional cardiology (16%). Limitations include the low survey response rate, which raises the risk of selection bias.

CONCLUSIONS This study provides insight into the current landscape of cardiac critical care delivery in North America, highlighting wide variation in staffing models, subspecialty training, and clinical practice. Our findings highlight growing trends toward high-intensity staffing models that incorporate critical care medicine-boarded physicians in consultative, comanagement, and or primary roles. (JACC Adv. 2026;5:102372) © 2025 The Authors. Published by Elsevier on behalf of the American College of Cardiology Foundation. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

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ABBREVIATIONS AND ACRONYMS

ACC = American College of Cardiology

AHA = American Heart Association

AHF = advanced heart failure

CCC = critical care cardiology

CICU = cardiac intensive care unit

C4 = Critical Care Cardiology Council

FTE = full-time equivalent

IC = interventional cardiology

ICU = intensive care unit

The purpose of the first coronary care unit, incepted by Dr Desmond Julian in 1961, was to provide electrocardiographic monitoring and early defibrillation to patients with myocardial infarction-related ventricular arrhythmias.[1] Technological advancements in diagnostic and therapeutic capabilities have transformed the spectrum and severity of disease encountered in the contemporary cardiac population, and by necessity Julian's coronary care unit has since evolved into the modern cardiac intensive care unit (CICU).[2-5] What was once predominantly an ischemic heart disease population has become a complex mixture of both acute and chronic cardiac and noncardiac etiologies of critical illness.[6] Currently, half of all CICU admissions are complicated by extracardiac organ dysfunction such as respiratory failure, renal failure, and sepsis, which are associated with increased mortality.[4,7,8]

The changing demographic of disease and marked increase in complexity has created the clinical need for cardiologists in the CICU to develop skills outside the scope of general cardiology or standard subspecialty cardiology training. This need has driven the emergence of critical care cardiology (CCC) as a distinct subspecialty, as recently codified by the American College of Cardiology (ACC) Critical Care Cardiology Council (C4) and American Heart Association (AHA).[5,9-11] In its early stages as a discipline, there is unsurprisingly significant heterogeneity in CCC training, competency, and practice patterns.[12-14] This variability is driven in part by differences in institutional needs, patient populations, procedural capabilities, and the scope of practice across local service lines. While the CCC physician pool is growing, the majority of CICUs are staffed by noninvasive cardiologists, heart failure cardiologists, and interventional cardiologists that leverage a variety of consultative and/or collaborative staffing approaches to incorporate physicians with needed critical care expertise.

As many centers have discovered, high-value CICU care is integral to successful patient outcomes and the programmatic growth of many of the medical and surgical cardiac subspecialties; however, how best to develop and sustain high-value care in the CICU remains uncertain. With this background in mind, the ACC CCC Member Section created a workforce survey to characterize the various existing approaches to CICU care delivery to guide the ongoing evolution of existing and emerging CICU programs and to further inform standards in CCC training.

METHODS

On behalf of the ACC C4 Workforce Work Group, we conducted a self-administered 42-item online multiple-choice survey of U.S. and Canadian CICU-focused cardiologists designed to capture the demographics of this workforce and the CICU operational models in which they practice. The survey underwent multiple iterations of development, was reviewed by ACC staff with expertise in survey design, and was pilot tested by both ACC staff and the physician investigators (the full survey is available online as a Supplemental Material). Sex, age, career stage, and practice location were assessed from the ACC database. The survey questions covered topics including CICU operational models (eg, patient population and staffing), CICU cardiologist training, practice characteristics, work expectations, and career satisfaction. A CICU cardiologist was defined as a cardiologist who functions as the primary managing clinician in a medical or surgical CICU. While we acknowledge that a subset of CICU-focused cardiologists have widespread CCC clinical experience and maintain sufficient CICU-based practice to be considered a critical care cardiologist,[11] for the purpose of this manuscript the term dual-boarded critical care cardiologist refers only to the subset of CICU cardiologists who were boarded or board eligible in critical care medicine. A high-intensity staffing model was defined as any model in which a dedicated, full-time critical care medicine-boarded physician directly managed or comanaged patient care (ie, comanagement between a critical care medicine-boarded physician and a cardiologist and or by a dual-boarded critical care cardiologist).

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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Advarra Institutional Review Board Services reviewed the study protocol, and using the Department of Health and Human Services regulations found at 45 CFR 46.104(d) determined that this research was exempt from Institutional Review Board oversight in the United States and Canada.

The survey link was emailed to 1,085 U.S. and Canadian CICU cardiologists from either the ACC database or contact information provided by the primary investigators. The survey was open from October 25, 2023, through March 4, 2024. The data have been presented as counts and percentages for categorical variables and median value with 25% to 75% IQR for continuous variables. Categorical variables were compared using chi-square tests when expected cell counts were sufficient; otherwise, Fisher exact test was applied. A P value <0.05 was considered statistically significant. A 2-tailed Spearman's correlation analysis was performed to assess the correlation between career satisfaction and various domains pertinent to the role of a CICU cardiologist. Missing responses were categorized and counted in descriptive totals but excluded from correlation analyses. The statistical analysis was performed using IBM SPSS Statistics for Windows Version 27.0.

RESULTS

A total of 218 physicians from more than 72 institutions completed the anonymous survey (20% response rate), and after excluding those still in training or not boarded or board eligible in cardiovascular medicine, a sample of n = 166 CICU cardiologists remained for the final analysis. The sample was 16% female, predominantly White (63%), and mainly between the ages of 30 to 50 years old (59%, Table 1). Respondents who self-identified as Asian (19%), Hispanic (8%), and Black (3%) represented a minority of the sample. Respondents most commonly reported working in large (>500 bed, 64%), university-affiliated academic medical centers (65%), followed by nonuniversity affiliated academic medical centers (17%), community hospitals (4%), and Veterans Affairs hospitals (1%). We observed that dual-boarded critical care cardiologists predominantly practiced in university-affiliated academic medical centers (77%), whereas nondual-boarded CICU cardiologists were more widely distributed across practice settings. Most of the respondents reported that their centers had durable left ventricular assist device programs (68%) and orthotopic heart transplant programs (60%). Similarly, most reported their centers initiate and manage intra-aortic balloon pumps (93%), microaxial percutaneous ventricular assist devices (86%), and extracorporeal membrane oxygenation (81%). Thus, the majority of the respondents were from centers with AHA Level 1 CICUs.[11]

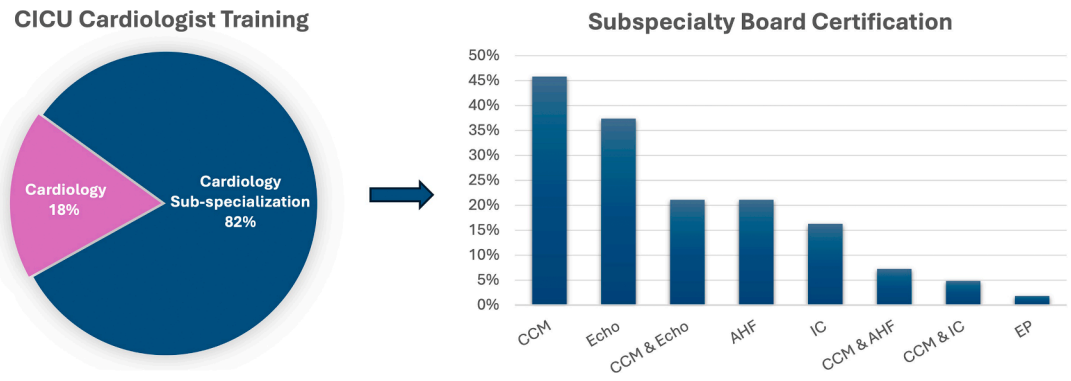
Cardiology subspecialization with board certification in critical care medicine was most common (46%), followed by board certification in echocardiography (37%), advanced heart failure (21%), and interventional cardiology (16%). A minority of the sample (18%) reported no subspecialization beyond

TABLE 1 CICU Cardiologist Demographics (N = 166)

Sex		
Male	118	71%
Female	27	16%
Unknown	21	13%
Race/Ethnicity		
White	104	63%
Black/African American	5	3%
Hispanic/Latino	14	8%
Asian	32	19%
Other	6	4%
Do not wish to disclose	5	3%
Age		
31-40 y	43	26%
41-50 y	54	33%
51-60 y	20	12%
61-70 y	23	14%
71 y and older	8	5%
Unknown	18	11%
Location		
United States	157	95%
Canada	9	5%
Hospital size (beds)		
Small (<100)	3	2%
Medium (100-500)	46	34%
Large (>500)	86	64%
Practice affiliation		
University-affiliated academic medical center	107	65%
Nonuniversity-associated academic medical center	28	17%
Community hospital	23	14%
VA or other government hospital	6	4%
Other	2	1%
Cardiac transplant and MCS capabilities of institution		
Initiate and manage IABP	155	93%
Initiate and manage microaxial percutaneous ventricular assist devices	143	86%
Initiate and manage ECMO	134	81%
Implant and manage durable left ventricular assist devices	112	68%
Perform heart transplantation	100	60%

CICU = cardiac intensive care unit; ECMO = extracorporeal membrane oxygenation; IABP = intra-aortic balloon pump; MCS = mechanical circulatory support; VA = Veterans Affairs.

FIGURE 1 CICU Cardiologist Training



Training background of the CICU cardiologist sample (n = 166). AHF = advanced heart failure; CCM = critical care medicine; CICU = cardiac intensive care unit; Echo = echocardiography; EP = electrophysiology; IC = interventional cardiology.

general cardiology (Figure 1). A third of the sample (33%) were dual-boarded in cardiology and critical care medicine and had additional certifications in either echocardiography (21%), advanced heart failure (7%), or interventional cardiology (5%).

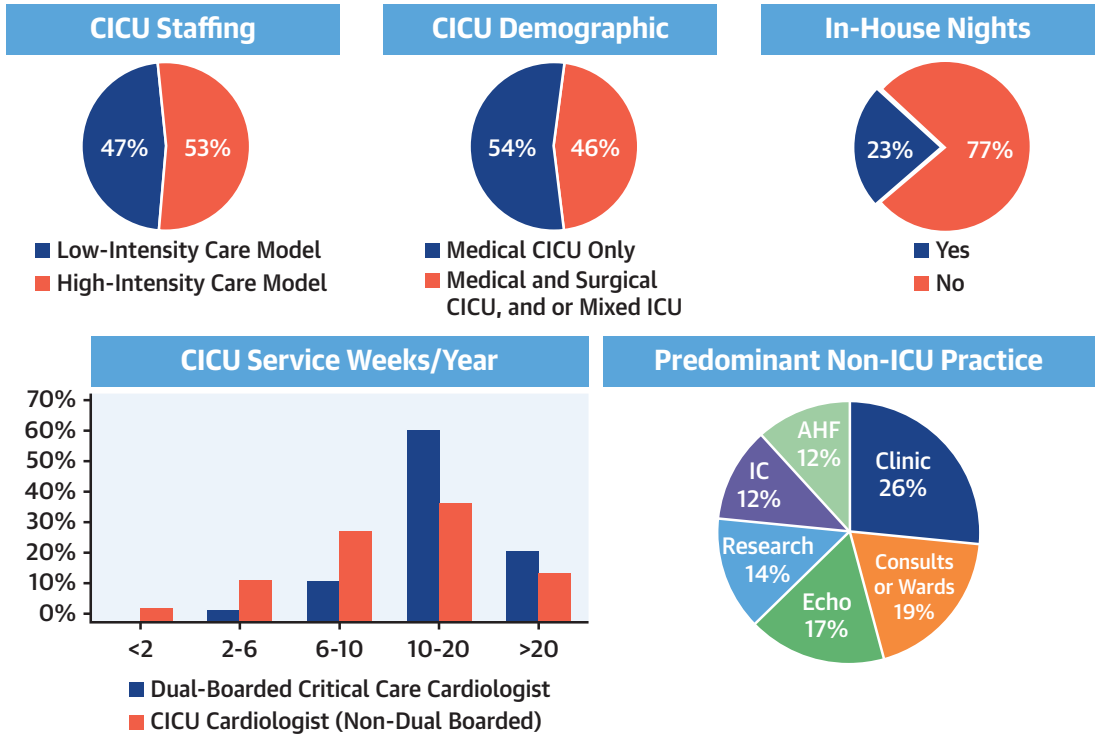
We asked the respondents to describe the operational structure and staffing models of the CICUs at their institutions. Fifty-three percent reported working in a CICU with a high-intensity staffing model (Central Illustration). More than half reported that their CICU was dedicated exclusively to the medical cardiology population (61%), whereas the remainder reported having a mixed medical and surgical cardiac population or an intensive care unit (ICU) model dedicated to all types of critically ill patients (cardiac and noncardiac). In respect to spectrum of practice, approximately half reported spending all of their ICU time in a medical CICU (54%), whereas the other half reported splitting time across some combination of the medical cardiology, cardiac surgery, and/or mixed cardiac/noncardiac ICU populations.

We observed 3 general approaches to incorporating critical care medicine-boarded physicians into CICU care delivery (Table 2). The sample reported that critical care medicine physicians were available as consultants (53%), through collaborative comanagement care models (29%), and/or the primary CICU attending was a dual-boarded critical care cardiologist (44%). The majority reported having traditional CICU-based house-staff rounding teams that included residents and fellows (78%), and half incorporated advanced practice providers (51%). Nocturnal in-house CICU staffing included residents and fellows (66%), advanced practice providers (37%), critical care medicine physicians (27%), CICU cardiologists (23%), and hospitalists (18%). More than two-thirds of CICU cardiologists reported providing nocturnal home-call (69%).

We observed that both the dual-boarded critical care cardiologists and nondual-boarded CICU cardiologists commonly perform typical cardiovascular procedures, including electrical cardioversion (97% vs 86%), venous and arterial vascular access (95% vs 73%), temporary transvenous pacemaker insertion (86% vs 71%), transesophageal echocardiography (71% vs 49%), and pericardiocentesis (54% vs 53%, Figure 2). Whereas the dual-boarded critical care cardiologists more commonly performed general critical care medicine procedures such as thoracentesis (83% vs 14%), endotracheal intubation (76% vs 16%), bronchoscopy (68% vs 0%), paracentesis (67% vs 14%), and chest tube insertion (63% vs 3%). Insertion of intra-aortic balloon pump (37% vs 39%), microaxial percutaneous ventricular assist device implantation (13% vs 22%), and extracorporeal membrane oxygenation cannulation (17% vs 14%) rates were similar among groups. Notably, 1% of the dual-boarded critical care cardiologists and 8% of nondual-boarded CICU cardiologists reported that they did not perform any of the queried procedures.

The median annual CICU service commitment of the sample was in the range of 10 to 20 weeks (IQR: 6-10 to 10-20). The median number of days or nights of in-house CICU coverage constituting a full-time equivalent (FTE) work expectation was 140 per year

CENTRAL ILLUSTRATION **Job Description of a CICU Cardiologist**



Substantial variation exists in the organization and delivery of cardiac critical care, as well as in the roles of CICU cardiologists across North America.

Papolos AI, et al. JACC Adv. 2026;5(1):102372.

Practice patterns and work expectation of the CICU cardiologist sample (n = 166). CICU service week work expectation stratified by dual-boarded critical care cardiologist (n = 76) and nondual-boarded CICU cardiologist subgroups (n = 90). Non-ICU practice patterns reported as predominate non-ICU work effort per respondent. ICU = intensive care unit; other abbreviations as in Figure 1.

(IQR: 90-180). Assuming that a week of CICU service equates to 7 days or nights, the median 100% CICU FTE work expectation was equal to 20 weeks of service annually, and the majority (65%) reported having at least 1 day free from clinical activities following a CICU service week.

Most CICU cardiologists reported that their job description included non-ICU service (97%). The predominant non-CICU work effort was most commonly outpatient cardiology clinic (25%), followed by general inpatient cardiology consults or wards (18%), echocardiography (16%), research (13%), interventional cardiology (11%), and advanced heart failure (11%).

Seventy percent of respondents reported being highly to extremely satisfied in their career. There was a moderately strong correlation between career satisfaction and satisfaction in the domains of work-related challenges regarding childcare and family duties (rho = 0.73; $P < 0.001$), amount of work requiring completion from home (rho = 0.68; $P < 0.001$), work-life balance (rho = 0.67; $P < 0.001$), and burnout (rho = −0.61; $P < 0.001$). Interestingly, we found no such correlation in relation to compensation (rho = −0.089; $P = 0.285$), working with trainees (rho = −0.070; $P = 0.379$), working in-house nights (rho = 0.070; $P = 0.376$), or total amount of CICU service worked per year (rho = −0.064; $P = 0.558$).

DISCUSSION

CICUs have become increasingly complex, with critically ill patients often suffering multi-organ

TABLE 2 CICU Staffing Models (N = 166)

Any hospital credentialed cardiologist can manage a patient they admit to the CICU (ie, open care model)	50	30%
A critical care medicine physician is available for consultation	88	53%
A critical care medicine physician manages or comanages patients as the primary provider	48	29%
A critical care medicine-boarded cardiologist manages patients as the primary provider	73	44%
Resident and/or fellow team members	130	78%
Advanced practice provider team members	84	51%
CICU nighttime staffing model		
In-house residents or fellow team members	110	66%
In-house advanced practice provider team members	62	37%
In-house critical care medicine physicians manage or comanage patients as the primary provider	43	26%
In-house hospitalists manage or comanage patients as the primary provider	29	18%
In-house CICU cardiologist as the primary provider	38	23%
Home-call CICU cardiologist available by phone	144	69%

Abbreviation as in Table 1.

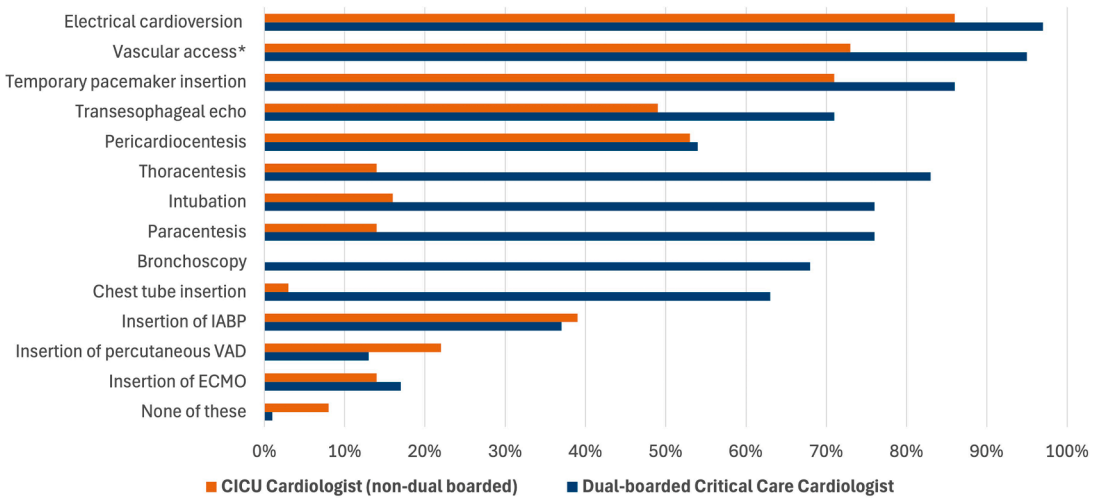
dysfunction requiring critical care medicine expertise. To meet this need, the burgeoning field of CCC has undergone rapid expansion in recent years, as evidenced by broadening institutional recognition and adoption, professional society representation, an emerging research infrastructure, and a growing number of CCC fellowship programs.[12] Utilizing results from the ACC C4 Workforce survey, we have presented new insight into modern CCC care delivery in North America. We observed notable variability across several facets of CICU organization and that the physician workforce is highly heterogeneous with respect to training, spectrum of practice, procedural competency, and work expectation.

The wide variability in operational organization and clinical practice provided by this report underscores 2 important points. First, determining which of the various CICU staffing models provides the highest value care would greatly serve patients and health care systems alike. While there is probably no "one-size-fits-all" solution, we suspect that there are tangible benefits to some approaches over others. The limited data available suggest that the high-intensity staffing model may be superior to the low-intensity model in terms of hospital mortality[15,16] and that staffing CICUs with dual-boarded critical care cardiologists may be advantageous.[17-19] A 2016 survey of academic medical centers from the AHA reported that 36% of CICUs had high-intensity staffing models, 24% were dedicated specifically to medical cardiology patients, and 25% had dual-boarded critical care cardiologists.[20] With the caveat that we surveyed CICU cardiologists and not institutions, it appears as though a shift in CICU care delivery is well underway, as we observed more than twice as many CICUs were dedicated to medical cardiology patients, and that CICUs were staffed by dual-boarded critical care cardiologists roughly twice as often. Additionally, we found that more centers were utilizing high-intensity staffing models, which was achieved through the incorporation of critical care medicine-boarded physicians via consultative, collaborative comanagement, and/or primary roles.

Second, CCC trainees require broad clinical exposure, as competencies and work expectations vary significantly across institutions. Notably, half of the respondents focus exclusively on medical cardiology patients, while the other half also practice in surgical CICUs. Therefore, to maximize potential career opportunities, it would be prudent for CCC trainees to develop the skills necessary to function effectively in mixed medical and surgical CICUs. Additionally, exposure to a variety of traditional cardiology and critical care medicine procedures is essential for practice as a critical care cardiologist in the contemporary CICU environment, along with the ability to manage various forms of temporary and durable mechanical circulatory support, as well as care for preoperative and postoperative heart transplant patients. Many of the dual-boarded critical care cardiologists in our sample reported routinely performing procedures such as endotracheal intubation, thoracentesis, and bronchoscopy—skills that demand dedicated training typically acquired through critical care medicine or CCC fellowships. Looking ahead, the field of CCC will need to establish structured curricula and standardized training guidelines, especially for trainees pursuing dual certification in cardiovascular and critical care medicine. This effort is actively underway in the form of a joint ACC/AHA Advanced Training Statement in CCC.

This survey provides preliminary evidence that despite working in a high-intensity environment with critically ill patients, most CICU cardiologists are satisfied in their career. We observed that key determinants predicting job satisfaction center around work-life balance and burnout. It is well known that cardiologists and critical care medicine physicians have high rates of burnout[21,22] and exploring models of care delivery that mitigate this issue could help grow the field. Our respondents described a wide variety of schedule structures including the number of total weeks of CICU time, distribution of other clinical and nonclinical duties, and number of consecutive CICU and recovery days.

FIGURE 2 CICU Cardiologist Procedural Practice



Active procedural competency of the dual-boarded critical care cardiologist (n = 76) and nondual-boarded CICU cardiologist (n = 90) subgroups. *Vascular access includes arterial lines, central lines, pulmonary artery catheters, and hemodialysis lines. ECMO = extracorporeal membrane oxygenation; IABP = intra-aortic balloon pump; VAD = ventricular assist device; other abbreviation as in Figure 1.

This implies that CICU cardiologists may have some flexibility in crafting their individual FTE allocation and perhaps optimizing one's schedule to respect work-life balance and/or periodically reallocating one's FTE distribution could serve as a potential deterrent to burnout. In addition to discovering how to keep the current CICU workforce in practice, these data highlight the need to train more CICU cardiologists from diverse gender and ethnic backgrounds, which is essential to providing equitable care, fostering creativity, and broadening research agendas to include a wider range of health issues.

We acknowledge that there are many aspects of CICU care that contribute to value beyond the scope of this analysis: hospital and patient demographics, nursing ratios and competency, temporary and durable mechanical support and transplant practice patterns, utilization of multidisciplinary care teams, and the availability and proficiency of ancillary staff including clinical pharmacists, respiratory therapists, registered dietitians, and physical therapists. A deeper dive into many of these facets of CICU care and their relation to patient outcomes is currently under investigation by the Critical Care Cardiology Trials Network and will hopefully shed more light on the subject.

Recognizing the benefits to both patient care and health care organizations, a growing number of prominent academic and community institutions have established CCC academic sections and clinical practices. Despite this momentum, significant barriers remain to structuring CCC roles effectively. These challenges often arise from administrative unfamiliarity with CICU staffing models, FTE expectations for CICU cardiologists, credentialing processes, and compensation structures. Because programmatic design is typically shaped by institutional priorities and resource availability, universal recommendations are difficult to prescribe. Nonetheless, we strongly urge institutional leaders to develop a deliberate, strategic framework—guided by insights from this survey and informed by consultation with experienced CCC professionals—to support the successful launch or expansion of CCC programs.

As many centers have discovered, the strategic growth of advanced heart failure programs and high-risk percutaneous coronary and structural intervention programs depend on patient success, procedural volume, and CICU capacity, each of which are intertwined with and reliant upon high-value CCC delivery. Here we have characterized the various existing approaches to operating and staffing a contemporary CICU with the aim to guide programs as they care for the changing needs of the critically ill cardiac population.

STUDY LIMITATIONS. The survey tool used for this study was novel and did not undergo formal pilot testing or validation. Due to the limited response rate and geographical distribution of the survey respondents, selection bias is likely and could preclude generalizability of our results especially in light of multiple respondents from some institutions. Several survey questions allowed nonmutually exclusive answers or used categorical range responses rather than quantitative values, precluding formal statistical analysis. Additionally, 81% of the respondents were from academic medical centers and thus our data may overestimate the prevalence of dual-boarded critical care cardiologists and use of advanced heart failure therapies and mechanical circulatory support. We recognize that comprehensive CICU care often involves collaboration across multiple specialties, including pulmonary-critical care medicine, anesthesiology-critical care, emergency medicine, and various surgical subspecialties. Our intention in this focused study was not to diminish the critical roles of these other specialists but rather to specifically characterize the evolving role, training, and practice patterns of cardiologists working in critical care settings, as defined by the ACC C4 Workforce survey.

CONCLUSIONS

The changing demographic and increasing complexity of disease in the modern CICU has led to the development of CICU-focused cardiologists and a restructuring of CICU care delivery. Here we have characterized the various existing approaches to providing CICU care, showing that there is significant heterogeneity in operations, practice patterns, and physician training. Our findings highlight growing trends toward high-intensity staffing models that incorporate critical care medicine-boarded physicians in consultative, comanagement, and/or primary roles. We recommend that institutional leaders adopt a structured approach to establishing and growing CCC programs incorporating data presented in this report. These data can help guide the evolving landscape of CCC as well as the development of best practices and standards in training.

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DISCLAIMER The views, information or content, and conclusions presented do not necessarily represent the official position or policy of, nor should any official endorsement be inferred on the part of, the Clinical Center, the National Institutes of Health, or the Department of Health and Human Services.

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APPENDIX For an expanded Methods section, please see the online version of this paper.